PBL Portal Content - Devansh Mathur

Smart Hospital Management System (SmartCare) | Team Nobita (PBL2627-AI&DS-A-031;) | Mentor: Er. Ram Babu Buri

Member	Devansh Mathur (24E1ARADM40P040)
Module	Module 1 - Authentication, Admin and Security
Code owned	LoginServlet, LogoutServlet, AuthService, PasswordUtil (BCrypt), AuthenticationFilter, AuthorizationFilter, AdminServlet, UserDAO, DepartmentDAO, AuditDAO, DBConnection, admin JSPs (dashboard, users, departments, audit), Docker + Render + cloud MySQL deployment.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

How to use this file

1) Daily Work Log tab: pick the Date, type Hours Worked, keep Status as shown, paste the description, click Submit Today's Work. 2) Weekly Report tab: choose the Week, paste the GitHub link, then paste the three boxes (Work Completed, Problems Faced, Plan for Next Week) from Part A. 3) Part A (weekly) comes first, Part B has the day-wise table. Submit your own entries only, from your own login.

PART A - Weekly Reports (13 weeks)

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Joined team of five, discussed project ideas and shortlisted a hospital management system. (2) Studied problems in manual hospital record keeping and listed login and role requirements. (3) Prepared the abstract section on authentication, roles and admin control. (4) Reviewed the guide's research area and confirmed Servlet-JSP with MySQL as the tech stack. (5) Finalised the abstract with the team and prepared it for portal submission.
Problems Faced	Difficulty in choosing one idea among several. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Identified five user roles: Admin, Doctor, Patient, Nurse and Receptionist. (2) Wrote functional requirements for login, logout, session handling and role-based access. (3) Wrote security requirements: password hashing, session timeout, audit trail.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Identified five user roles: Admin, Doctor, Patient, Nurse and Receptionist. (2) Wrote functional requirements for login, logout, session handling and role-based access. (3) Wrote security requirements: password hashing, session timeout, audit trail. (4) Documented non-functional requirements: performance, availability and usability. (5) Compiled the authentication and admin part of the SRS and reviewed it with the team.
Problems Faced	Unclear which roles needed separate dashboards. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for the Admin actor (manage users, departments, audit logs). (2) Drew the login and logout sequence diagram with filters and session. (3) Designed class diagram for User, Role and AuditLog.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Drew the use-case diagram for the Admin actor (manage users, departments, audit logs). (2) Drew the login and logout sequence diagram with filters and session. (3) Designed class diagram for User, Role and AuditLog. (4) Designed class diagram for Department and its relation with Doctor. (5) Merged the module's UML with the team diagrams and fixed relationships.
Problems Faced	Mixing class diagram relations for User and Role. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed ER tables roles, users and audit_logs with keys and constraints. (2) Designed the departments table and linked doctors to departments. (3) Created Maven project structure with pom.xml and Jakarta Servlet dependencies.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Designed ER tables roles, users and audit_logs with keys and constraints. (2) Designed the departments table and linked doctors to departments. (3) Created Maven project structure with pom.xml and Jakarta Servlet dependencies. (4) Created the GitHub repository and shared the basic package layout with the team. (5) Prepared login page and admin dashboard mock-ups.
Problems Faced	Foreign key order created errors in SQL script. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the Role and User model classes with getters and setters. (2) Created DBConnection utility reading db.properties. (3) Implemented UserDAO methods to fetch user by username and email.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote the Role and User model classes with getters and setters. (2) Created DBConnection utility reading db.properties. (3) Implemented UserDAO methods to fetch user by username and email. (4) Added UserDAO insert, update and delete operations with PreparedStatement. (5) Implemented DepartmentDAO CRUD and tested queries on local MySQL.
Problems Faced	Driver not found error for MySQL connector. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote PasswordUtil using the jBCrypt library for hashing and verification. (2) Built AuthService login flow with credential and role checking. (3) Implemented LoginServlet doGet and doPost with redirect by role.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote PasswordUtil using the jBCrypt library for hashing and verification. (2) Built AuthService login flow with credential and role checking. (3) Implemented LoginServlet doGet and doPost with redirect by role. (4) Implemented LogoutServlet to invalidate the session safely. (5) Wrote AuditDAO to record login, logout and admin actions.
Problems Faced	Role redirect looped back to login page. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created login.jsp with responsive form and error messages. (2) Created admin dashboard.jsp with summary cards. (3) Built admin users.jsp with table for add, edit and delete users.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Created login.jsp with responsive form and error messages. (2) Created admin dashboard.jsp with summary cards. (3) Built admin users.jsp with table for add, edit and delete users. (4) Built admin departments.jsp and audit.jsp views. (5) Added common header, navbar and sidebar includes and linked style.css.
Problems Faced	Include files broke relative paths in admin pages. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote ValidationUtil for email, phone and empty field checks. (2) Created ApplicationException and DatabaseException classes. (3) Implemented AuthenticationFilter to block pages for logged-out users.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote ValidationUtil for email, phone and empty field checks. (2) Created ApplicationException and DatabaseException classes. (3) Implemented AuthenticationFilter to block pages for logged-out users. (4) Implemented AuthorizationFilter for role-based URL protection. (5) Configured session timeout, cookie settings and error pages in web.xml.
Problems Faced	Filter blocked static CSS and JS files. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Completed AdminServlet actions for user and department management. (2) Added duplicate username and email validation in user creation. (3) Added activate and deactivate user option for admin.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Completed AdminServlet actions for user and department management. (2) Added duplicate username and email validation in user creation. (3) Added activate and deactivate user option for admin. (4) Reviewed module code, removed unused code and added comments. (5) Pushed module code to GitHub and updated the tracking sheet.
Problems Faced	Merge conflicts in shared files. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Merged all modules into one WAR project and resolved package conflicts. (2) Connected the login redirect to every role dashboard. (3) Checked filter rules against every servlet URL pattern.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Merged all modules into one WAR project and resolved package conflicts. (2) Connected the login redirect to every role dashboard. (3) Checked filter rules against every servlet URL pattern. (4) Fixed navbar and sidebar links that differed by role. (5) Ran full-flow login to logout across all five roles.
Problems Faced	Sidebar links showed wrong items for some roles. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Prepared test cases for login, wrong password and locked access. (2) Tested role-based access by opening restricted URLs directly. (3) Tested session expiry and logout behaviour.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Prepared test cases for login, wrong password and locked access. (2) Tested role-based access by opening restricted URLs directly. (3) Tested session expiry and logout behaviour. (4) Tested validation on user and department forms. (5) Logged defects found in testing and fixed critical ones.
Problems Faced	Session lost after server restart during tests. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote a Dockerfile (Maven build stage and Tomcat 10.1 run stage). (2) Created a cloud MySQL instance and moved the schema to it. (3) Updated db.properties for the cloud database with SSL.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote a Dockerfile (Maven build stage and Tomcat 10.1 run stage). (2) Created a cloud MySQL instance and moved the schema to it. (3) Updated db.properties for the cloud database with SSL. (4) Deployed the WAR to Render and checked the live site. (5) Fixed password comparison and role matching issues for demo accounts.
Problems Faced	Cloud DB SSL connection refused at first. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the introduction and tools sections of the project report. (2) Documented authentication and security design in the report. (3) Prepared slides on architecture, roles and security.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote the introduction and tools sections of the project report. (2) Documented authentication and security design in the report. (3) Prepared slides on architecture, roles and security. (4) Recorded and edited the demo video of login and admin features. (5) Practised viva questions on filters, sessions and BCrypt.
Problems Faced	Video recording took multiple attempts. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

PART B - Daily Work Log (Monday to Friday, 65 entries)

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description
06-07-2026 (Mon)	2	Completed	Joined team of five, discussed project ideas and shortlisted a hospital management system.
07-07-2026 (Tue)	1.5	Completed	Studied problems in manual hospital record keeping and listed login and role requirements.
08-07-2026 (Wed)	2	Completed	Prepared the abstract section on authentication, roles and admin control.
09-07-2026 (Thu)	1.75	Completed	Reviewed the guide's research area and confirmed Servlet-JSP with MySQL as the tech stack.
10-07-2026 (Fri)	2	Completed	Finalised the abstract with the team and prepared it for portal submission.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description
13-07-2026 (Mon)	1.75	Completed	Identified five user roles: Admin, Doctor, Patient, Nurse and Receptionist.
14-07-2026 (Tue)	1.75	Completed	Wrote functional requirements for login, logout, session handling and role-based access.
15-07-2026 (Wed)	1.5	Completed	Wrote security requirements: password hashing, session timeout, audit trail.
16-07-2026 (Thu)	1.75	Completed	Documented non-functional requirements: performance, availability and usability.
17-07-2026 (Fri)	1.5	Completed	Compiled the authentication and admin part of the SRS and reviewed it with the team.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description
20-07-2026 (Mon)	1.5	Completed	Drew the use-case diagram for the Admin actor (manage users, departments, audit logs).
21-07-2026 (Tue)	2	Completed	Drew the login and logout sequence diagram with filters and session.
22-07-2026 (Wed)	1.75	Completed	Designed class diagram for User, Role and AuditLog.
23-07-2026 (Thu)	1.5	Completed	Designed class diagram for Department and its relation with Doctor.
24-07-2026 (Fri)	2	Completed	Merged the module's UML with the team diagrams and fixed relationships.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description
27-07-2026 (Mon)	1.5	Completed	Designed ER tables roles, users and audit_logs with keys and constraints.
28-07-2026 (Tue)	1.75	Completed	Designed the departments table and linked doctors to departments.
29-07-2026 (Wed)	1.75	Completed	Created Maven project structure with pom.xml and Jakarta Servlet dependencies.
30-07-2026 (Thu)	1.75	Completed	Created the GitHub repository and shared the basic package layout with the team.
31-07-2026 (Fri)	1.5	Completed	Prepared login page and admin dashboard mock-ups.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description
03-08-2026 (Mon)	1.5	Completed	Wrote the Role and User model classes with getters and setters.
04-08-2026 (Tue)	1.5	Completed	Created DBConnection utility reading db.properties.
05-08-2026 (Wed)	2	Completed	Implemented UserDAO methods to fetch user by username and email.
06-08-2026 (Thu)	1.75	Completed	Added UserDAO insert, update and delete operations with PreparedStatement.
07-08-2026 (Fri)	1.75	Completed	Implemented DepartmentDAO CRUD and tested queries on local MySQL.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description
10-08-2026 (Mon)	2	Completed	Wrote PasswordUtil using the jBCrypt library for hashing and verification.
11-08-2026 (Tue)	1.75	Completed	Built AuthService login flow with credential and role checking.
12-08-2026 (Wed)	1.5	Completed	Implemented LoginServlet doGet and doPost with redirect by role.
13-08-2026 (Thu)	1.75	Completed	Implemented LogoutServlet to invalidate the session safely.
14-08-2026 (Fri)	1.5	Completed	Wrote AuditDAO to record login, logout and admin actions.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description
17-08-2026 (Mon)	2	Completed	Created login.jsp with responsive form and error messages.
18-08-2026 (Tue)	1.5	Completed	Created admin dashboard.jsp with summary cards.
19-08-2026 (Wed)	2	Completed	Built admin users.jsp with table for add, edit and delete users.
20-08-2026 (Thu)	2	Completed	Built admin departments.jsp and audit.jsp views.
21-08-2026 (Fri)	2	Completed	Added common header, navbar and sidebar includes and linked style.css.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description
24-08-2026 (Mon)	1.75	Completed	Wrote ValidationUtil for email, phone and empty field checks.
25-08-2026 (Tue)	1.5	Completed	Created ApplicationException and DatabaseException classes.
26-08-2026 (Wed)	2	Completed	Implemented AuthenticationFilter to block pages for logged-out users.
27-08-2026 (Thu)	2	Completed	Implemented AuthorizationFilter for role-based URL protection.
28-08-2026 (Fri)	1.75	Completed	Configured session timeout, cookie settings and error pages in web.xml.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description
31-08-2026 (Mon)	1.5	Completed	Completed AdminServlet actions for user and department management.
01-09-2026 (Tue)	2	Completed	Added duplicate username and email validation in user creation.
02-09-2026 (Wed)	2	Completed	Added activate and deactivate user option for admin.
03-09-2026 (Thu)	2	Completed	Reviewed module code, removed unused code and added comments.
04-09-2026 (Fri)	1.5	Completed	Pushed module code to GitHub and updated the tracking sheet.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description
07-09-2026 (Mon)	2	Completed	Merged all modules into one WAR project and resolved package conflicts.
08-09-2026 (Tue)	1.75	Completed	Connected the login redirect to every role dashboard.
09-09-2026 (Wed)	2	Completed	Checked filter rules against every servlet URL pattern.
10-09-2026 (Thu)	2	Completed	Fixed navbar and sidebar links that differed by role.
11-09-2026 (Fri)	1.5	Completed	Ran full-flow login to logout across all five roles.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description
14-09-2026 (Mon)	1.75	Completed	Prepared test cases for login, wrong password and locked access.
15-09-2026 (Tue)	2	Completed	Tested role-based access by opening restricted URLs directly.
16-09-2026 (Wed)	1.75	Completed	Tested session expiry and logout behaviour.
17-09-2026 (Thu)	2	Completed	Tested validation on user and department forms.
18-09-2026 (Fri)	1.75	Completed	Logged defects found in testing and fixed critical ones.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description
21-09-2026 (Mon)	1.75	Completed	Wrote a Dockerfile (Maven build stage and Tomcat 10.1 run stage).
22-09-2026 (Tue)	1.75	Completed	Created a cloud MySQL instance and moved the schema to it.
23-09-2026 (Wed)	1.75	Completed	Updated db.properties for the cloud database with SSL.
24-09-2026 (Thu)	2	Completed	Deployed the WAR to Render and checked the live site.
25-09-2026 (Fri)	1.5	Completed	Fixed password comparison and role matching issues for demo accounts.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description
28-09-2026 (Mon)	2	Completed	Wrote the introduction and tools sections of the project report.
29-09-2026 (Tue)	2	Completed	Documented authentication and security design in the report.
30-09-2026 (Wed)	2	Completed	Prepared slides on architecture, roles and security.
01-10-2026 (Thu)	1.75	Completed	Recorded and edited the demo video of login and admin features.
02-10-2026 (Fri)	1.75	Completed	Practised viva questions on filters, sessions and BCrypt.

Total hours in this log: 115.75 over 65 days.